Divisor of Algebraic Curves (Foundations Ver.)

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Description of this Presentation

- The Topic of this Final Paper is "Divisors of Algebraic Curves".
- The foundations of this topic are Classical Algebraic Geometry, Scheme Theory and Commutative Algebra.
- While the key machinaries of the topic are Sheaf Theory, Invertible Sheaves, and Weil/Cartier Divisors.
- Hence in this presentation, we shall present some foundational topics.

Motivation

- Given polynomials $p_1, \ldots, p_r \in k[X_1, \ldots, X_n]$, we wish to understand the geometry of the solutions of $p_1 = \cdots = p_r = 0$.
- For example, $p(x, y) = x^2 + y^2 - 1$ cuts out a circle: one equation in the plane gives a *curve*.
- On a curve we can ask for the zeros and poles of a rational function, *together with their multiplicities*.
- For example, $f = \dfrac{x-1}{x^2}$ on $\mathbb{A}^1_k$ has a simple zero at 1 and a double pole at 0. We record this as the formal sum $[1] - 2[0]$.
- Such formal sums of points are called divisors. To make this precise, we need rings, topology and sheaves.

Roadmap

1. **Noetherian rings, Zariski topology, Nullstellensatz:** the dictionary between algebra and geometry.
2. **Affine varieties:** coordinate rings, polynomial maps, rational functions, local rings.
3. **Sheaves and affine schemes:** gluing local data; Spec A and its structure sheaf.
4. **Looking ahead:** Weil divisors, Cartier divisors, invertible sheaves and Pic (the main body of the paper).

Outline of this section

1. Noetherian Rings
2. Zariski Topology
3. Nullstellensatz Theorem

Noetherian Rings

Definition 1 (Noetherian Ring)

A ring A is *Noetherian* if the following equivalent conditions hold:

(i) Every ideal $I \subset A$ is finitely generated; that is, for every ideal $I \subset A$, there exist $f_1, \ldots, f_k \in I$ such that $I = (f_1, \ldots, f_k)$.

(ii) Every ascending chain

$$I_1 \subset I_2 \subset \cdots \subset I_m \subset \ldots$$

of ideals of A terminates, that is, the chain is eventually stationary, with $I_N = I_{N+1} = \ldots$ (the ascending chain condition, or a.c.c.).

(iii) Every nonempty set of ideals of A has a maximal element.

- For example, $\mathbb{Z}$ and $k[x]$ are Noetherian.
- The intuition of Noetherian Ring is we want to work in ring that has finite generators.

Proposition 2

(i) Suppose that A is Noetherian, and $I \subset A$ an ideal; then the quotient ring $B = A/I$ is Noetherian.
(ii) Let A be a Noetherian integral domain, and $A \subset K$ its field of fractions; let $\emptyset \neq S \subset A$ be a subset, and set

$$B = A[S^{-1}] = \left\{ \frac{a}{b} \in K \,\middle|\, a \in A,\ b = 1 \text{ or a product of elements of } S \right\}.$$

Then B is again Noetherian.

Theorem 3 (Hilbert Basis Theorem)

For a ring A,

$$A \text{ Noetherian} \Rightarrow A[X] \text{ Noetherian}.$$

Corollary 4

For k a field, a finitely generated k-algebra is Noetherian.

Algebraic set

Definition 5

A subset $X \subset \mathbb{A}_k^n$ is an *algebraic set* if $X = V(I)$ for some ideal I. Notice that by Corollary 3.3, I is finitely generated. If $I = (f_1, \ldots, f_r)$ then clearly

$$V(I) = \{P \in \mathbb{A}_k^n \mid f_i(P) = 0 \text{ for } i = 1, \ldots, r\},$$

so that an algebraic set is just a locus of points satisfying a finite number of polynomial equations.

Zariski Topology

Proposition 6 (Proposition-Definition)

The correspondence V satisfies the following formal properties:

(i) $V(0) = \mathbb{A}_k^n$; $\quad V(A) = \emptyset$;

(ii) $I \subset J \implies V(J) \subset V(I)$;

(iii) $V(I_1 \cap I_2) = V(I_1) \cup V(I_2)$;

(iv) $V\left(\sum_i I_i\right) = \bigcap_i V(I_i)$.

Hence the algebraic subsets of $\mathbb{A}_k^n$ form the closed sets of a topology on $\mathbb{A}_k^n$, the *Zariski topology*.

V-I Correspondence

Let $A = k[X_1, \ldots, X_n]$.

$$\{\text{ideals } J \subset A\} \longrightarrow \{\text{subsets } X \subset \mathbb{A}_k^n\}$$
$$J \mapsto V(J) = \{P \in \mathbb{A}_k^n \mid f(P) = 0 \text{ for all } f \in J\}.$$

As a kind of inverse to V there is a correspondence

$$\{\text{ideals } J \subset A\} \longleftarrow \{\text{subsets } X \subset \mathbb{A}_k^n\}$$
$$I(X) = \{f \in k[X_1, \ldots, X_n] \mid f(P) = 0 \text{ for all } P \in X\}. \hookleftarrow X$$

That is, I takes a subset X to the ideal of functions vanishing on it.

Radical Ideal

- Before stating the Nullstellensatz, we define the radical of an ideal in order to simplify the statement.

Definition 7 (Radical ideal)

Let I be an ideal of $A = k[X_1, \ldots, X_n]$. The **radical** of I is

$$\operatorname{rad} I = \{f \in A : f^m \in I \text{ for some integer } m \geq 1\}.$$

If $I = \operatorname{rad} I$, we say that I is a **radical ideal**.

- $\operatorname{rad} I$ is an ideal containing I, and $V(\operatorname{rad} I) = V(I)$.
- Every prime ideal is radical, and $I(X)$ is a radical ideal for every $X \subseteq k^n$.
- For example, in $k[X]$ we have $\operatorname{rad}(X^2) = (X)$. Both ideals define the same algebraic set $\{0\}$, so the algebraic set does not remember the multiplicity.

Hilbert's Nullstellensatz

Theorem 8 (Nullstellensatz, weak form)

Let k be an algebraically closed field and $A = k[X_1, \ldots, X_n]$. If $\mathfrak{m}$ is a maximal ideal of A, then

$$\mathfrak{m} = (X_1 - a_1,\ X_2 - a_2,\ \ldots,\ X_n - a_n)$$

for some point $(a_1, a_2, \ldots, a_n) \in k^n$.

Theorem 9 (Nullstellensatz, strong form)

Let k be an algebraically closed field and $A = k[X_1, \ldots, X_n]$. If J is an ideal of A with $J \neq A$, then $V(J) \neq \emptyset$. Furthermore, for every ideal J of A,

$$I(V(J)) = \operatorname{rad} J.$$

The Algebra–Geometry Dictionary

Corollary 10

The maps $X \mapsto I(X)$ and $J \mapsto V(J)$ are mutually inverse, inclusion-reversing bijections:

Geometry (in k^n)		**Algebra** (in $A = k[X_1, \ldots, X_n]$)
algebraic sets	$\longleftrightarrow$	radical ideals
irreducible	$\longleftrightarrow$	prime ideals
points	$\longleftrightarrow$	maximal ideals

- So the geometry of k^n can be recovered from the ring A alone.
- **Idea of scheme theory:** replace A by an arbitrary commutative ring with 1, and take *all* prime ideals as the points of the space.

Outline of this section

Polynomial Maps

Definition 11

Polynomial Maps Define $f : V \to W$ as polynomial maps, given by m polynomial functions $f_1, \ldots, f_m \in k[V]$,, so that for $P \in V$,

$$f(P) = \big(f_1(P), \ldots, f_m(P)\big) \in W.$$

- Keep in mind that if we have composite of f with any polynomial function $g \in k[W]$ is again polynomial function on V.
- We can see that polynomial maps evaluate each coordinates by its polynomial functions, hence we have such proposition.

Proposition 12

Suppose we have $V \subset \mathbb{A}_k^n$, $W \subset \mathbb{A}_k^m$, and we write $f = (f_1, \ldots, f_m)$, then

$$f : V \to W \text{ is polynomial map } \iff \forall j, f_j = Y_j \circ f \in k[W]$$

by given Y_j j-th coordinate function.

- Given polynomial map $f : V \to W$, now we define map

$$f^* : k[W] \to k[V]$$
$$g \mapsto g \circ f$$

- Then f^* is ring homomorphism, and in fact, it is k-algebra homomorphism.
- Note that it went backwards: from coordinate ring of the target to that of the source.
- Now with that, we propose a claim

Proposition 13

Every k-algebra homomorphism $\Phi\colon k[W] \to k[V]$ is of the form $\Phi = f^*$ for a *uniquely defined* polynomial map $f\colon V \to W$.

- Before defining affine variety, we want the following claim.

Proposition 14

Given $V \subset \mathbb{A}_k^n$, we have

$$V \text{ is irreducible} \iff I(V) \text{ prime ideal} \iff k[V] \text{ is integral domain}$$

Sketch proof. We shall prove the contrapositive X reducible $\leftrightarrow$ $I(X)$ is not prime. If X is reducible, then we have $X = X_1 \cup X_2$ and choose $f_1 = I(X_1)\backslash I(X)$ and $f_2 = I(X_2)\backslash I(X)$, hence $f_1, f_2 \notin I(X)$ but $f_1 f_2 \in I(X)$ since it vanishes there. If the ideal is not prime, then $\exists f_1, f_2 \notin I(X)$ but $f_1 f_2 \in I(X)$. Construct $I_1 = (I(X), f_1)$ and $X_1 = V(I_1)$, while samewise with $X_2 = V(I_2)$, then $X = X_1 \cup X_2$. The second iff is obvious since $k[V] = k[X_1, \ldots, X_n]/I(V)$.

- We want V be irreducible, so that we can use $k[V]$ to define a quotient field, which will be shown later.

Affine Varieties

Definition 15 (Affine varieties)

An *affine variety* over a field k is an irreducible algebraic subset $V \subset \mathbb{A}^n_k$, equipped with its coordinate ring $k[V]$. Equivalently, it is a set V together with a ring $k[V]$ of k-valued functions $f : V \to k$ such that:

1. $k[V]$ is a finitely generated k-algebra, and
2. for some choice $x_1, \ldots, x_n$ of generators of $k[V]$ over k, the map

$$V \to \mathbb{A}^n_k, \qquad P \mapsto \big(x_1(P), \ldots, x_n(P)\big)$$

embeds V as an irreducible algebraic set.

Morphism between Affine Varieties

- Now we wish to know the mapping between affine varieties.
- Intuitively, we would use usual mapping consists of polynomial function, as previous shown.
- However, it is too restrictive as we can have rational function.
- For example, suppose we have Pythogoras equation $X^2 + Y^2 = Z^2$ and we wish to transform it into circle by $x = X/Z$ and $y = Y/Z$, so we have $x^2 + y^2 = 1$ so finally we can have parametrisation of $\lambda = \frac{x}{1-y}$, hence $x = \frac{2\lambda}{\lambda^2+1}$ and $y = \frac{\lambda^2-1}{\lambda^2+1}$
- And notice that using the mapping of $f(x) = 1/x$ is not polynomial, hence we need to define rational mapping.

Rational Function

- Given V is irreducible and $k[V]$ domain integral, then define

$$k(V) := \text{Quot}(k[V])$$

  as $\text{Quot}(R)$ is quotient field of domain integral R.
- By this, we can write $f \in k(V)$ as $f = g/h$ for $h \neq 0$ (in the sense of zero polynomial).
- But notice f is not well defined if $\exists P \in V$ such that $h(P) = 0$.
- Hence we define as following

Definition 16

Let $f \in k(V)$ and $P \in V$, define f is *regular* at P if $h(P) \neq 0$ with $f = g/h, g, h \in k[V]$, and define *domain of definition* of f by

$$\text{dom} f := \{P \in V \mid f \text{ is regular at } P\}$$

and define *local ring* of V at P as

$$\mathcal{O}_{V,P} := \{f \in k(V) \mid f \text{ is regular at } P\}$$

- We seek the characteristics of the $\operatorname{dom} f$, define $D_f := \{h \in k[V] \mid hf \in k[V]\}$, then

$$V \setminus \operatorname{dom} f = \{P \in V \mid h(P) = 0\} = V(D_f)$$

  then $\operatorname{dom} f$ open in Zariski Topology. Since it is not empty, then it is dense.
- $\operatorname{dom} f = V$ if and only if $f \in k[V]$, i.e. f is a polynomial function, since

$$\operatorname{dom} f = V \iff V(D_f) = \emptyset \iff 1 \in D_f \iff f \in k[V]$$

- For any $h \in k[V]$, let $V_h = V \setminus V(h) = \{P \in V \mid h(P) \neq 0\}$. Then

$$\operatorname{dom} f \supset V_h \iff f \in k[V][h^{-1}].$$

  since

  $\operatorname{dom} f \supset V_h \iff h$ vanishes on $D_f \iff h^n \in D_f$ for some n $\iff f = g/h^n \in k[V][h^{-1}]$

- Hence, we can write the local ring as such

$$\mathcal{O}_{V,P} = k[V][(h^{-1} \mid h(P) \neq 0)]$$

- Finally, we can define rational map

Definition 17

A *rational map* $f : V \dashrightarrow \mathbb{A}_k^m$ is a partially defined map given by rational functions $f_1, \ldots, f_n \in k(V)$, that is,

$$f(P) = \big(f_1(P), \ldots, f_m(P)\big) \quad \text{for all } P \in \bigcap_i \operatorname{dom} f_i.$$

By definition, $\operatorname{dom} f = \bigcap_i \operatorname{dom} f_i$. f is said to be *regular* at $P \in V$ if and only if $P \in \operatorname{dom} f$. A rational map $f : V \dashrightarrow W$ between two affine varieties $V \subset \mathbb{A}_k^n$ and $W \subset \mathbb{A}_k^m$ is defined to be a rational map $f : V \dashrightarrow \mathbb{A}_k^m$ such that $f(\operatorname{dom} f) \subset W$.

- For example, consider $C = V(x^2 + y^2 - 1)$. The maps

$$\mathbb{A}^1 \dashrightarrow C,\ \lambda \mapsto \left(\frac{2\lambda}{\lambda^2+1}, \frac{\lambda^2-1}{\lambda^2+1}\right), \qquad C \dashrightarrow \mathbb{A}^1,\ (x,y) \mapsto \frac{x}{1-y}$$

  although the maps are not polynomial (the first is undefined where $\lambda^2 + 1 = 0$, the second where $y = 1$).

Outline of this section

Why Sheaves?

- Now we back to Nullstellensatz, where the idea is to replace the A by an arbitrary commutative ring with 1, and take *all* prime ideals as the points of the space.
- Before that, we need to define several objects.

Definition 18 (Presheaf)

Let X be a topological space. A *presheaf* $\mathcal{F}$ of rings on X assigns

- to each open $U \subset X$ a ring $\mathcal{F}(U)$ (the *sections* over U), and
- to each inclusion $V \subset U$ a ring homomorphism $\rho_{UV} : \mathcal{F}(U) \to \mathcal{F}(V)$, written $s \mapsto s|_V$,

such that $\mathcal{F}(\emptyset) = 0$, $\rho_{UU} = \mathrm{id}$, and $\rho_{VW} \circ \rho_{UV} = \rho_{UW}$ for $W \subset V \subset U$.

Sheaf Axioms

- Then we shall begin to define sheaf

Definition 19 (Sheaf)

A presheaf $\mathcal{F}$ on X is a *sheaf* if for every open U and every open cover $U = \bigcup_i U_i$:

(S1) *(Locality)* if $s \in \mathcal{F}(U)$ and $s|_{U_i} = 0$ for all i, then $s = 0$;

(S2) *(Gluing)* if $s_i \in \mathcal{F}(U_i)$ satisfy $s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$ for all i, j, then there is $s \in \mathcal{F}(U)$ with $s|_{U_i} = s_i$ for all i.

(By S1, the section s in S2 is unique.)

- For examples, Continuous real functions on a topological space X form a sheaf.
- And related example, for an affine variety V, $U \mapsto \mathcal{O}_V(U)$ is a sheaf, the *sheaf of regular functions*.
- For non-example, the *constant* presheaf $U \mapsto \mathbb{R}$ is not a sheaf: if $U = U_1 \sqcup U_2$ is disconnected, the sections 0 on U_1 and 1 on U_2 agree on the (empty) overlap but do not glue to a constant on U.

Stalks

Definition 20 (Stalk)

For a presheaf $\mathcal{F}$ on X and $x \in X$, the *stalk* at x is

$$\mathcal{F}_x := \varinjlim_{U \ni x} \mathcal{F}(U),$$

that is, germs of sections: $s \in \mathcal{F}(U)$ and $t \in \mathcal{F}(U')$ define the same germ iff $s|_W = t|_W$ for some open $W \ni x$ inside $U \cap U'$.

- For the sheaf $\mathcal{O}_V$ of an affine variety, the stalk at $P \in V$ is exactly the local ring $\mathcal{O}_{V,P}$ defined earlier.
- Stalks are where we do "local" algebra; they are what will let us define ord_P on a curve.

The Spectrum of a Ring

Definition 21 (Spec A)

Let A be a commutative ring with 1. As a set,

$$\operatorname{Spec} A := \{\mathfrak{p} \subset A \mid \mathfrak{p} \text{ is a prime ideal}\}.$$

For an ideal $I \subset A$ put $V(I) = \{\mathfrak{p} \in \operatorname{Spec} A \mid \mathfrak{p} \supseteq I\}$.

- $V(I)$ satisfies the same four properties as in the Zariski Topology slide, so the $V(I)$ are the closed sets of a topology on $\operatorname{Spec} A$ (the *Zariski topology*).
- For $f \in A$, the *principal open sets*

$$D(f) := \operatorname{Spec} A \setminus V(f) = \{\mathfrak{p} \mid f \notin \mathfrak{p}\}$$

  form a basis of the topology.
- $\{\mathfrak{p}\}$ is closed iff $\mathfrak{p}$ is maximal; in general $\overline{\{\mathfrak{p}\}} = V(\mathfrak{p})$. If A is a domain, (0) is a *generic point*: $\overline{\{(0)\}} = \operatorname{Spec} A$.

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